

GOLCUK DUMLUPINAR, TurkeyWMO: **170670**Lat: **40.7267N**Lon: **29.8067E**Elev: **59**StdP: **14.66**Time Zone: **3.00 (E03)**Period: **94-16**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 30.0	(c) 32.2	(d) 24.2	(e) 18.4	(f) 33.7	(g) 26.7	(h) 20.7	(i) 36.3	(j) 17.9	(k) 52.9	(l) 15.8	(m) 51.8	(n) 4.0	(o) 140	(p) 0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 18.5	(c) 93.3	(d) 73.9	(e) 90.6	(f) 73.1	(g) 88.2	(h) 72.2	(i) 77.1	(j) 88.7	(k) 75.6	(l) 86.5	(m) 74.2	(n) 84.6	(o) 5.5	(p) 90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.5	125.1	83.9	72.1	119.1	82.4	70.8	113.7	81.0	40.6	88.9	39.1	86.5	37.8	84.3	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 14.3	(b) 12.1	(c) 10.4	(e) 26.3	(f) 100.0	(g) 3.7	(h) 3.3	(i) 23.7	(j) 102.4	(k) 21.5	(l) 104.3	(m) 19.4	(n) 106.1	(o) 16.7	(p) 108.5		
			DB													
			WB													
				25.1	79.0	3.5	2.0	22.5	80.4	20.4	81.5	18.5	82.6	15.9	84.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.7	44.3	45.8	49.2	56.8	65.6	73.4	78.1	78.6	71.5	62.6	53.8	47.5		
	DBStd	13.63	6.92	7.32	6.71	6.66	5.70	4.80	3.89	4.48	5.43	6.22	6.64	6.81		
	HDD50	642	206	157	99	15	0	0	0	0	0	2	34	129		
	HDD65	3004	643	539	491	255	62	2	0	0	11	119	339	543		
	CDD50	4547	29	39	74	221	485	701	871	887	645	392	150	53		
	CDD65	1432	0	1	1	11	82	253	406	422	206	44	4	2		
	CDH74	11577	1	3	8	89	554	1970	3635	3796	1300	190	26	5		
	CDH80	4338	0	0	1	19	131	688	1486	1595	385	32	1	0		
(14) Wind	WSAvg	4.2	4.1	4.6	4.9	4.7	4.3	4.1	4.2	4.2	3.9	3.6	3.6	3.9		
(15) Precipitation	PrecAvg	28.5	3.2	2.6	2.7	2.2	1.8	1.9	1.2	1.4	2.1	3.2	2.8	3.6		
	PrecMax	40.9	6.6	5.4	4.4	5.2	6.0	4.9	3.5	6.9	4.4	7.5	5.7	8.9		
	PrecMin	19.7	1.0	0.4	0.9	0.3	0.2	0.1	0.0	0.0	0.0	0.2	0.3	1.4		
	PrecStd	4.8	1.5	1.3	0.9	1.2	1.3	1.2	1.1	1.7	1.4	1.8	1.5	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.9	70.6	73.9	82.5	88.7	94.1	96.4	97.4	91.3	83.6	78.0	72.7		
		MCWB	53.2	54.8	56.2	63.9	68.4	73.0	73.8	75.7	71.3	67.5	59.9	56.3		
	2%	DB	62.4	64.9	67.4	75.8	83.5	90.0	92.5	93.6	87.3	78.9	71.0	65.3		
		MCWB	51.2	53.1	54.3	60.7	66.7	71.7	73.5	74.7	70.5	65.1	59.0	54.4		
	5%	DB	58.8	60.5	63.7	71.4	79.7	86.8	90.0	90.8	84.4	75.2	66.8	61.5		
		MCWB	50.0	51.2	52.8	58.9	65.7	70.7	73.2	74.0	69.6	64.3	57.6	52.8		
	10%	DB	54.4	56.5	60.0	67.9	76.6	83.9	87.7	88.1	81.5	71.8	63.3	57.5		
		MCWB	48.5	49.6	51.3	57.4	64.5	69.5	72.5	72.9	68.6	63.4	56.6	51.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.8	57.0	57.5	65.1	70.7	76.4	78.6	79.2	75.4	69.9	63.0	58.1		
		MCDB	64.4	66.5	67.7	79.0	84.3	89.5	91.6	91.7	86.4	78.8	71.1	67.4		
	2%	WB	52.4	54.1	55.3	62.0	68.3	73.8	76.7	77.6	72.9	67.5	60.7	55.6		
		MCDB	60.3	61.7	65.7	73.6	80.1	85.6	88.4	89.2	82.4	75.1	67.6	62.4		
	5%	WB	50.5	52.0	53.6	59.8	66.7	72.3	75.2	76.2	71.3	65.6	58.9	53.9		
		MCDB	57.0	59.4	62.0	69.8	77.7	83.4	86.5	86.9	80.5	72.5	64.8	59.9		
	10%	WB	48.8	50.0	51.9	57.9	65.1	70.8	73.9	74.7	69.8	64.0	57.0	51.9		
		MCDB	54.0	56.0	58.9	66.9	74.9	81.4	84.3	84.6	78.8	70.2	62.0	56.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	11.1	12.4	14.2	16.6	17.7	18.5	18.6	18.5	17.6	13.8	12.9	11.1		
		MCDBR	17.5	19.1	21.4	23.3	23.5	23.6	22.4	22.3	22.0	19.7	18.4	16.0		
		MCWBR	8.9	9.8	10.5	11.5	11.3	10.5	9.2	8.5	9.2	9.1	9.2	8.1		
	5% WB	MCDBR	16.2	17.6	19.3	21.5	21.0	20.6	19.1	19.7	18.6	16.4	15.7	13.7		
		MCWBR	9.0	9.5	9.8	11.4	10.8	10.1	9.0	8.5	8.4	8.3	8.4	7.5		
(40) Clear-Sky Solar Irradiance	taub		0.361	0.390	0.431	0.472	0.441	0.428	0.431	0.437	0.425	0.405	0.372	0.361		
	taud		2.333	2.234	2.136	2.042	2.205	2.288	2.294	2.273	2.293	2.325	2.361	2.349		
	Ebn at Noon		246	255	258	255	267	270	268	262	257	248	240	236		
	Edh at Noon		28	36	44	52	45	42	41	41	37	33	27	26		
(44) All-Sky Solar Radiation	RadAvg		493	724	1108	1520	1909	2177	2253	1969	1472	973	656	460		
	RadStd		71	106	106	173	170	192	86	127	117	87	84	69		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.19	N/A	N/A	+2.12	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (13 neighbors)		+0.86	N/A	N/A	N/A	N/A	N/A	N/A	-162	N/A	N/A

Nomenclature: See separate page